

Reading Data from Pictures

Tutorial

MAL3040 — Statistics (B.Sc. B.Ed.)

How to use this tutorial

Each exhibit is shown twice: first the picture on its own, then the summary numbers together with some questions.

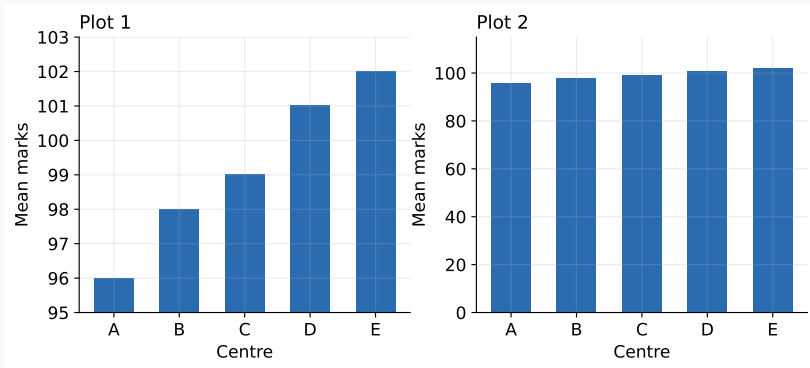
1. On the first slide, say in words what you see: where the centre lies, how spread out the data are, whether the shape is symmetric, whether anything looks unusual.
2. Only then look at the numbers, and ask whether they match your impression.
3. Where picture and numbers disagree, the interesting question is *why*.

No calculation is required. Every answer is a sentence, not a number.

Recipe. Every data set here is simulated. The rule used to generate it is given at the foot of the slide on which it first appears, so that nothing is hidden from you.

Part 1 — The picture is a choice

Two pictures of the same five numbers



Recipe. The fixed values 96, 98, 99, 101, 102, plotted twice. The only difference between the panels is the range of the vertical axis: 95–103 on the left, 0–115 on the right.

Two pictures of the same five numbers

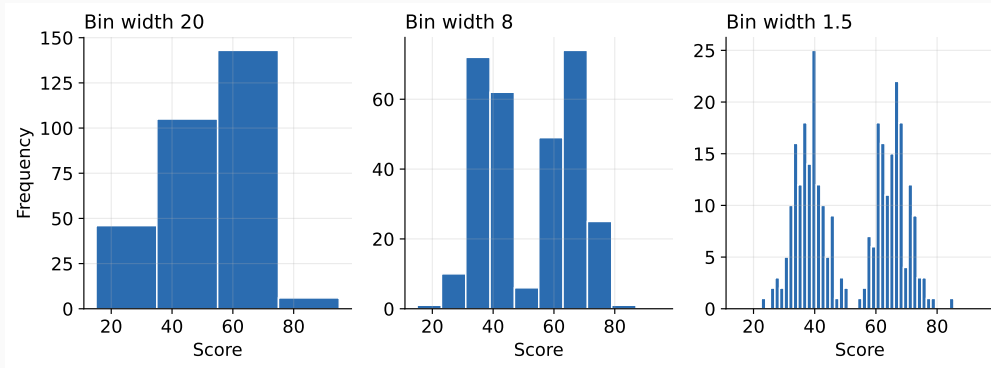
Mean marks at five centres:

A	B	C	D	E
96	98	99	101	102

Questions

- From Plot 1, how large is the gap between Centre A and Centre E? And from Plot 2?
- Which plot would a newspaper print in order to report a scandal? Which one to report that all is well?
- Is either plot *false*? What exactly changed between them?
- When, if ever, is it legitimate to start the vertical axis away from zero?

The same 300 observations, three histograms



Recipe. 300 values: 150 drawn from a normal distribution with mean 38 and standard deviation 5.5, and 150 drawn from a normal distribution with mean 66 and the same standard deviation. Nothing changes across the three panels except the bin width of the histogram.

The same 300 observations, three histograms

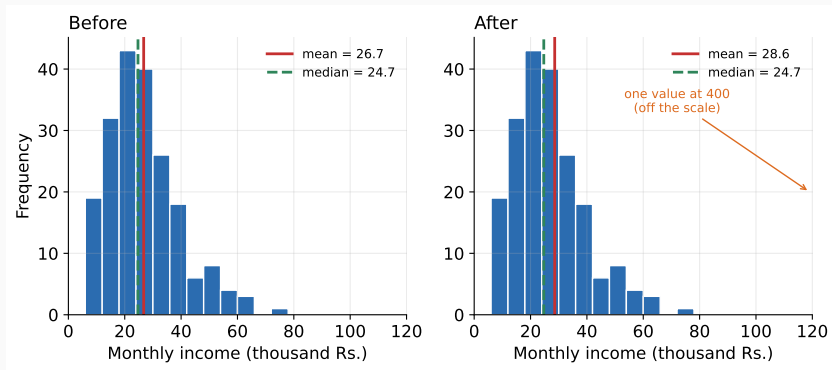
$n = 300$, $\bar{x} = 51.89$, $s = 14.84$ — identical for all three panels.

Questions

- How many groups does the class appear to fall into, according to each panel?
- The mean is about 52. How many students actually scored near 52?
- Which panel would you trust, and how would you decide?
- What does this exhibit say about reporting only $\bar{x}$ and s ?

Part 2 — Centre and spread

One new observation



Recipe. 200 values drawn from a lognormal distribution with median 24 and log-scale spread 0.45, rounded to one decimal place; the single value 400 is then appended to give the second panel.

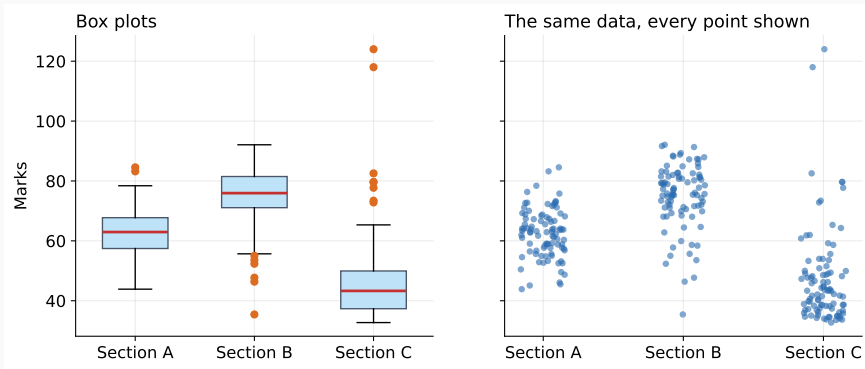
One new observation

	n	$\bar{x}$	median	s	CV
Before	200	26.72	24.70	12.53	46.9%
After	201	28.58	24.70	29.15	102.0%

Questions

- One household earning Rs. 4,00,000 a month has been added to a survey of 200 households. Which summaries moved a great deal, and which hardly moved at all?
- “The average monthly income here is about Rs. 28,600.” Is that true? Is it honest?
- Which single number would you report to a village council, and why?
- Why does the standard deviation react so much more violently than the mean?

Reading a box plot



Recipe. Section A: normal, mean 62, standard deviation 8. Section B: 95 minus a gamma variable of shape 4 and scale 5. Section C: 30 plus a gamma variable of shape 2.2 and scale 7, with the two values 118 and 124 appended. Here $n = 90$ per group, and 92 in Section C.

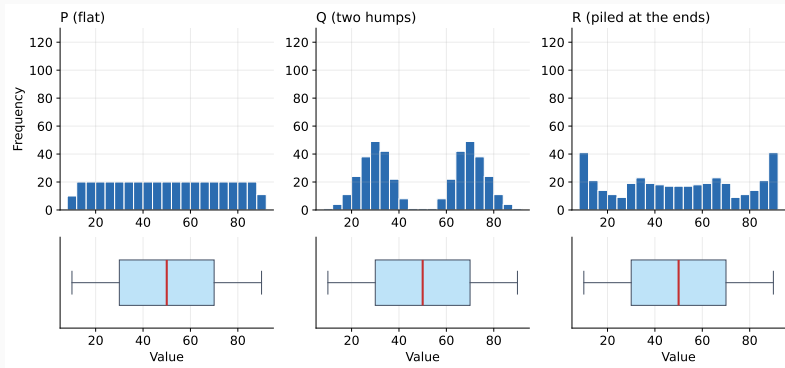
Reading a box plot

Group	n	$\bar{x}$	median	Q_1	Q_3	IQR	s
Section A	90	62.50	62.95	57.44	67.74	10.30	8.14
Section B	90	74.94	75.94	71.08	81.49	10.41	10.71
Section C	92	47.35	43.29	37.34	49.94	12.61	15.96

Questions

- Which section is symmetric? Which is stretched towards low marks, and which towards high marks?
- In Section B the mean lies below the median; in Section C it lies above. Explain both facts from the shape.
- The orange points lie beyond $Q_1 - 1.5 \text{ IQR}$ or $Q_3 + 1.5 \text{ IQR}$. Does that make them mistakes?
- A and B have nearly equal IQRs (10.30, 10.41) but standard deviations 8.14 and 10.71. How?

Three data sets with identical box plots



Recipe. 401 points taken as evenly spaced quantiles of three densities — a uniform density, a two-bump mixture, and a U-shaped density — each then passed through an increasing piecewise-linear map that sends its own five-number summary onto 10, 30, 50, 70, 90. An increasing map does not reorder the data, so the quartiles land exactly where intended.

Three data sets with identical box plots

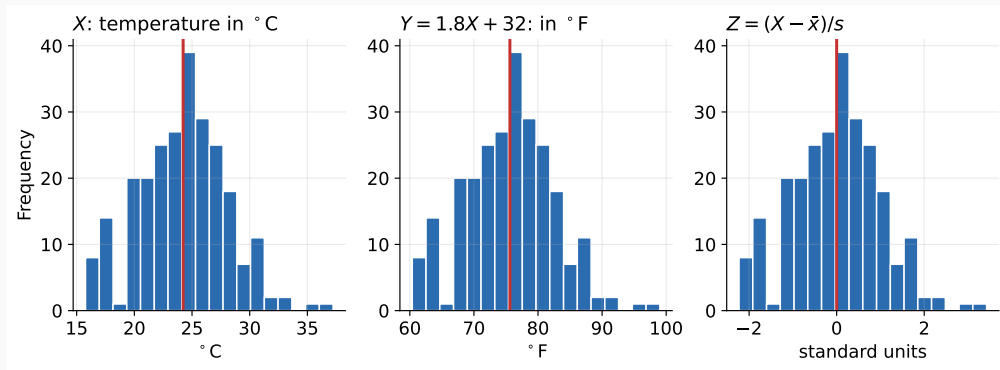
Set	n	min	Q_1	median	Q_3	max	$\bar{x}$	s
P (flat)	401	10	30	50	70	90	50.00	23.18
Q (two humps)	401	10	30	50	70	90	50.00	21.52
R (piled at the ends)	401	10	30	50	70	90	50.00	26.20

Questions

- All three share the same minimum, quartiles, median, maximum *and* mean. From the box plots alone, could you have told them apart?
- Each set has exactly 100 observations in each of 10–30, 30–50, 50–70 and 70–90. Why must that be so?
- Could any data set with this five-number summary have a single peak in the middle?
- So when is a box plot the wrong picture to draw?

Part 3 — Changing the units

$$X \mapsto aX + b$$



Recipe. 250 values drawn from a normal distribution with mean 24 and standard deviation 4, read as degrees Celsius. The remaining two panels show the very same numbers after $y = 1.8x + 32$ and after $z = (x - \bar{x})/s$.

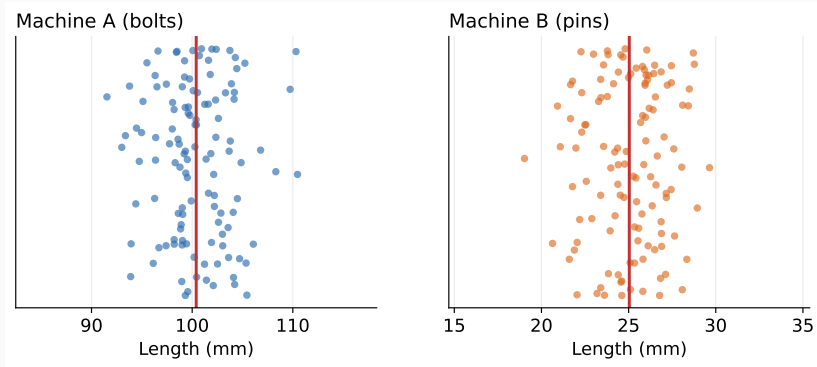
$$X \mapsto aX + b$$

	$\bar{y}$	median	s	CV
X ($^{\circ}\text{C}$)	24.229	24.425	3.795	15.66%
$Y = 1.8X + 32$ ($^{\circ}\text{F}$)	75.613	75.966	6.831	9.03%
$Z = (X - \bar{x})/s$	-0.000	0.052	1.000	—

Questions

- Compare the three histograms. What has changed, and what has not?
- Check in the table that $\bar{y} = 1.8\bar{x} + 32$ and $s_y = 1.8s_x$. Would $s_y = 1.8s_x$ still hold if a were -1.8 ?
- The coefficient of variation falls from 15.66% to 9.03%, and yet the weather has not changed. What has gone wrong?
- What are the mean and standard deviation of Z , and why?

Which machine is more consistent?



Recipe. Machine A: 120 values from a normal distribution with mean 100 and standard deviation 4. Machine B: 120 values from a normal distribution with mean 25 and standard deviation 2. Each panel is drawn on its own horizontal scale, five standard deviations either side of its own mean.

Which machine is more consistent?

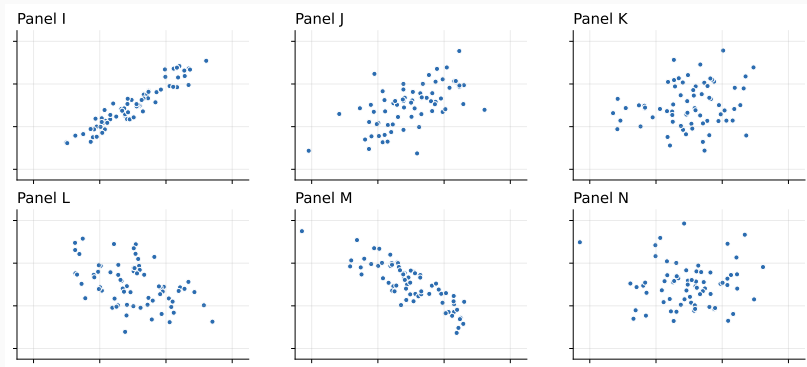
	n	$\bar{x}$ (mm)	s (mm)	CV
Machine A	120	100.40	3.59	3.57%
Machine B	120	25.04	2.07	8.27%

Questions

- The two clouds look about equally spread. Look at the axes — why is that appearance misleading?
- By standard deviation, Machine A varies more; by coefficient of variation, Machine B varies more. Which is the fair comparison here, and why?
- On the previous exhibit the CV of a temperature changed when the units changed. Why is the CV trustworthy here but not there?
- What property must a variable have before its CV means anything?

Part 4 — Association between two variables

Guess the correlation



Recipe. Each panel holds 70 independent standard normal pairs, affinely remapped so that $\bar{x} = \bar{y} = 50$, $s_x = s_y = 10$, and r takes a chosen value exactly. The axis tick labels are deliberately hidden. Write down your guess for each panel before moving on.

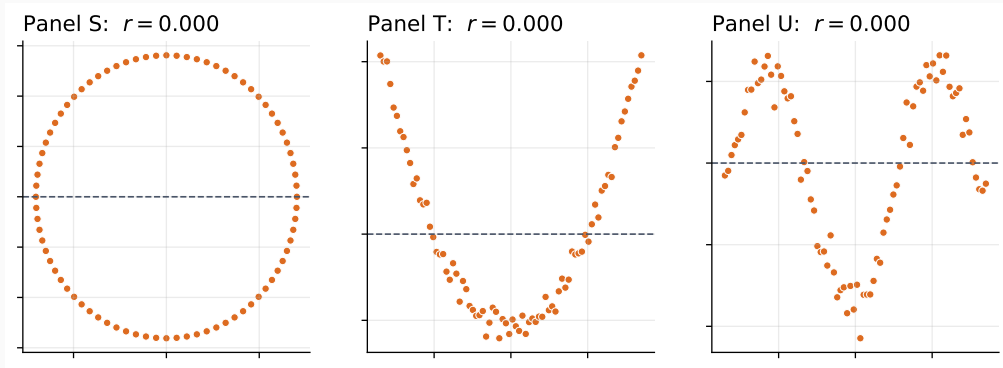
Guess the correlation — answers

Panel I	Panel J	Panel K	Panel L	Panel M	Panel N
$r = 0.95$	$r = 0.60$	$r = 0.20$	$r = -0.45$	$r = -0.85$	$r = 0.00$

Questions

- How close were your guesses? Most people underestimate $|r|$ in the middle of the range.
- Order the six panels from strongest to weakest association. Does that ordering agree with the ordering by $|r|$?
- Panels K and N look much alike but have $r = 0.20$ and $r = 0.00$. Could you have separated them by eye?
- Does a small value of $|r|$ mean that x and y are unrelated?

Three data sets with $r = 0$



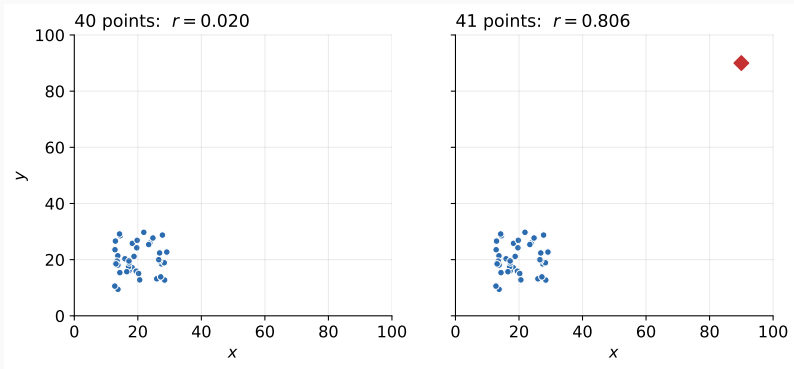
Recipe. Panel S: 80 points equally spaced on a circle. Panel T: 80 points on $y = x^2$ with small noise. Panel U: 80 points on $y = \sin(1.6x)$ with small noise. Each is then affinely remapped so that r is exactly 0.

Three data sets with $r = 0$

Questions

- In each panel, could you predict y reasonably well if you were told x ?
- So what exactly is r measuring, given that it equals 0 in all three cases?
- A report states: “we found no correlation between fertiliser dose and yield, so the dose does not matter”. What would you ask to see?
- Suggest a way of detecting the structure in Panel T from the numbers alone.

One point



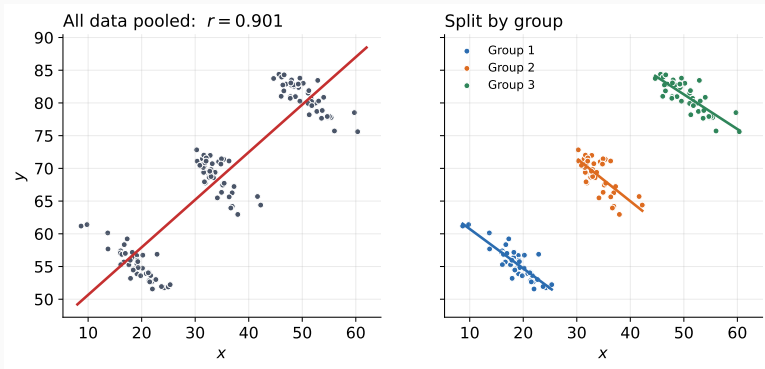
Recipe. 40 points drawn uniformly and then remapped to have $\bar{x} = \bar{y} = 20$, $s_x = s_y = 5.5$ and $r = 0.02$ exactly; the single point (90, 90) is then appended.

$$r_{40} = 0.020 \qquad r_{41} = 0.806$$

Questions

- The forty original points are untouched. Adding one observation moved r from 0.02 to 0.81. How is that possible?
- If you were handed only the number $r = 0.81$, what would you conclude?
- Would the *median* of x have moved as much? Why not?
- What should you always do before quoting a correlation coefficient?

Pooling three groups



Recipe. Three groups of 45 points, centred at $(20, 55)$, $(35, 68)$ and $(50, 81)$, each generated with a within-group slope of -0.55 and noise of standard deviation 1.6.

Pooling three groups

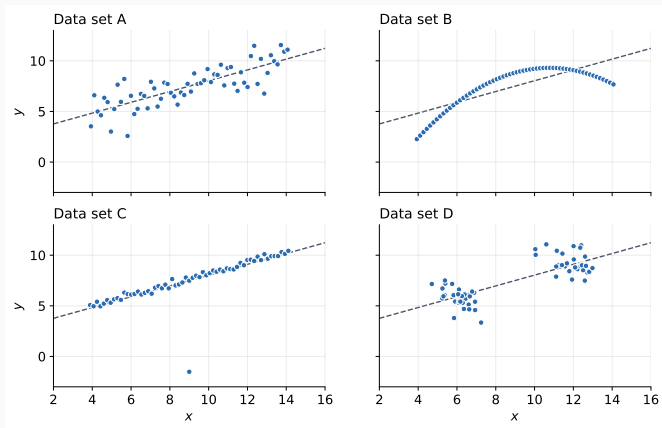
$$\text{pooled } r = 0.901 \quad r_1 = -0.855 \quad r_2 = -0.705 \quad r_3 = -0.824$$

Questions

- Read x as hours of private tuition per week, y as marks, and the three groups as three schools. Pooled, more tuition goes with higher marks; within every school, with *lower* marks. Are both true?
- Which of the two statements should guide a parent choosing tuition?
- What is it about the three schools that produces the reversal?
- What would you need to know before drawing any causal conclusion at all?

Part 5 — Putting it together

Four data sets

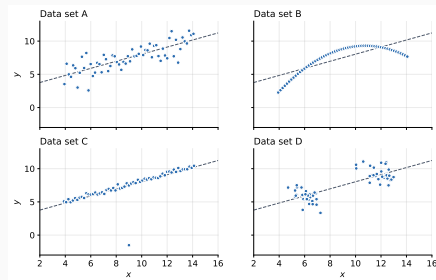


Recipe. Four sets of 60 points — a noisy straight line, a curve, a tight line with one distant point, and two clusters — each affinely remapped to the same $\bar{x}$, s_x , $\bar{y}$, s_y and r . The dashed line is identical in all four panels.

Four data sets — the summary numbers

Set	$\bar{x}$	s_x	$\bar{y}$	s_y	r
A	9.00	3.00	7.50	2.00	0.800
B	9.00	3.00	7.50	2.00	0.800
C	9.00	3.00	7.50	2.00	0.800
D	9.00	3.00	7.50	2.00	0.800

All four therefore share the same least-squares line, drawn dashed on each panel.



Questions

- A report hands you only this table. What have you not been told?
- In set C, what happens to r if the single low point is deleted? Guess the direction, and justify it.
- In set D, is it sensible to speak of a relationship between x and y at all?

What to take away

- A summary number is a *deliberate* loss of information. Know what you have thrown away.
- The mean and the standard deviation are pulled about by extreme observations; the median and the interquartile range are not.
- A box plot shows position and spread, not shape. A histogram shows shape, but the bin width is your choice.
- Under $y = ax + b$: the shape is unchanged, $\bar{y} = a\bar{x} + b$, $s_y = |a| s_x$, and r is unchanged when $a > 0$. The coefficient of variation is *not* preserved, since b moves $\bar{x}$ but leaves s alone.
- r measures *straight-line* association only, and a single observation can dominate it.
- Draw the picture first.

Notes for discussion (1/2)

Axis truncation Neither plot lies. Plot 1 makes a 6% difference look sixfold. The real question is which baseline is meaningful: for marks out of 100 a zero baseline is natural, for a share price it is not.

Bin width Width 20 conceals the two groups completely; width 1.5 displays sampling noise as though it were structure. The mean of about 52 describes almost nobody in the class.

Mean against median The median is unchanged at 24.70; the mean rises by about 1.86; and s more than doubles, from 12.53 to 29.15. The mean is a balance point, so a distant weight moves it; the squaring in s makes it more sensitive still.

Box plots A mean below the median indicates a left tail, a mean above it a right tail. The IQR ignores the tails and s does not, which is how Sections A and B have nearly equal IQRs while s differs by about a third: B's long left tail inflates s and leaves the quartiles almost untouched. The 1.5 IQR rule flags points for attention; it does not certify them as errors.

Identical box plots Since the five-number summary is equally spaced, each quarter of the data occupies an interval of equal width, so the counts per interval are *forced* to be equal, and only the shape within a quarter can vary. It follows that no data set with this five-number summary can have a single central peak — a good illustration of exactly what a box plot discards.

Notes for discussion (2/2)

Linear change The shape survives because the map is increasing and affine. $CV = s/\bar{x}$ is unaffected by scaling but not by a shift, so it is meaningful only for a variable with a genuine zero — length, weight, income — and not for a temperature in °C or °F. Note also $s_y = |a|s_x$, so a negative a flips the picture but leaves the spread alone.

Coefficient of variation Each machine is drawn on its own scale, which is precisely why the two clouds look alike. In millimetres $s_A > s_B$, but relative to the size of the object being made, Machine B is the less consistent of the two.

Guessing r Students typically guess too low for $|r|$ between 0.3 and 0.7; panels K and N are close to indistinguishable by eye.

$r = 0$ Correlation measures linear association. A circle, a parabola and a sine curve are all strongly structured and yet have $r = 0$ exactly. A plot of residuals against x would expose the curvature in Panel T at once.

One point The added observation lies far from both means, so its product of deviations dominates the numerator of r . The median of x would shift by a single position only.

Pooling Both statements are true of their own data sets. The reversal arises because the schools differ in *both* variables at once; the within-school comparison is the relevant one, and the pooled figure is confounded by school.

On the data

Every data set in this tutorial is synthetic, generated in Python with `numpy` under the fixed seed 20260817. The generating rule appears at the foot of the slide on which the data first appear.

Three of the exhibits use one common device. Given paired data (u, v) , let z_1 be the standardised u , and let z_2 be the standardised residual of v after its component along z_1 is removed. Then z_1 and z_2 have sample mean 0, sample variance 1, and sample correlation 0, so

$$x = \bar{x}_* + s_{x*} z_1, \quad y = \bar{y}_* + s_{y*} \left(r_* z_1 + \sqrt{1 - r_*^2} z_2 \right)$$

has exactly the prescribed $\bar{x}_*, s_{x*}, \bar{y}_*, s_{y*}, r_*$, while the qualitative shape of the scatter survives, the map being affine. This is how the four panels of the quartet, the six panels of the correlation grid, and the three $r = 0$ exhibits were constructed.